1. Use composite trapezoidal rule ,with $n = 4$ (number of subintervals) , to approximate the value of the integral $\int_1^3 \cos(\frac{3}{x})\,dx$, round off your your answer to 3 decimal places ?

A. 0.193 B. 1.335 C. 1.260 D. 0.381

2. (Gauss-Seidel method)

Using Gauss-Seidel method , to solve the linear system $x = \frac{7-y}{4}$, $y = \frac{-2-x}{5}$, $x_0 = 0$ *and* $y_0 = 0$ (don't change the equations) , then $y_1 =$

A. -0.75 B. -0.7 C. -0.95 D. -1

3. Using least squares regression to fit the data $(x_1, y_1), (x_2, y_2), \cdots, (x_{10}, y_{10})$ by the linear model $y_i = a_0 + a_1 x_i + e_i$ it gives $\sum_{i=1}^{10} x_i = 100$, $\sum_{i=1}^{10} x_i^2 = 200$, $\sum_{i=1}^{10} y_i = 80$ and $\sum_{i=1}^{10} x_i y_i = 500$ then $a_0 =$

A. 1 B. 3 C. 0.375 D. 4.25

4. Solving the equation , $f(x) = 0$, by Newton-Raphson method , if $x_0 = 2$, $f'(x_0) = 4$ and $f(x_0) = 2$ then $x_1 =$

A. -2 B. 0 C. 1.5 D. 1.75

5. Solving the intial value problem $\frac{dy}{dx} = 2x - \ln y$, $y(1) = 2$ by Euler's method , taking $h = 0.1$ as a step size . Approximate $y(1.2)$,rounding the answer to 4 decimal places?

A. 2.7556 B. 2.5713 C. 2.3576 D. 2.5712

6. The linear equation $a^2 x - 4x + 2 = a$, has unique solution when ;

A. $a = 2$ only B. $a = -2$ only C. $a = 2$ *or* $a = -2$ D. $a \neq 2$ *and* $a \neq -2$

7. By applying simple Simpson's $\frac{1}{3}$ – rule , approximate the integral , $\int_0^{\frac{\pi}{6}} \sqrt{\sin x}\,dx$?

(round off the answer to 4 decimal places)

A. 0.2393 B. 0.6561 C. 0.2501 D. 0.6866

8. Given that $f(x) = x^2$, $x_0 = 2$, $x_1 = 3$ *and* $x_2 = 5$, then the second divided difference $f[x_2, x_1, x_0] =$

A. 2.5 B. 2 C. 1 D. 1.5

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9. Using Modified Newton-Raphson method, to find the multiple root of the equation ,
$x^3 - 4x^2 + 4x = 0$ .If $x_0 = 1.7$ then $x_2 = ?$ (round off your answer to 4 decimal places)

A. 1.8234 B. 1.9699 C. 1.9693 D. 1.9998

10. If $f(x) = x^2 e^{-x} \sin x$, approximate $f'(1.6)$, using forward divided difference formula , to 5 decimal places , taking 0.02 as a step size?

A. 0.10510 B. 0.19375 C. 0.50718 D. 0.01290

11. Applying simple Simpson's $\frac{3}{8} - rule$,approximate the value of the integral , $\int_0^1 3\sec x \, dx$, to 5 decimalplaces?

A. 4.92145 B. 3.69109 C. 2.00010 D. 2.46073

12. Given that $x_0 = 1$, $x_1 = 2$ *and* $x_2 = 4$, if the second order Lagrange interpolation related to $f(x)$ is $f_2(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2)$ then $L_0(1.5) =$

A. $\frac{5}{12}$ B. $\frac{1}{3}$ C. $\frac{1}{3}$ D. $-\frac{5}{12}$

13. Given that $f(5) = f(5.0) + 2 = f(4.9) + 3 = 4$,use centered finite difference formula , approximate , $f''(5)$?

A. 500 B. 50 C. -50 D. -500

14. The graph of ; $f(x) = (x^2 - 1)(x - 1)^2 (x^3 + x^2)$

A. touches the $x-axis$ at $x = 1$.
B. crosses the $x-axis$ at $x = 1$.
C. touches the $x-axis$ at $x = 2$.
D. crosses the $x-axis$ at $x = -1$.

15. If a is a constant real number , then the solution of the system , $A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = B$, whose augumented matrix;

$$\left[\begin{array}{ccc:c} 1 & 2 & -1 & -3a \\ 0 & 1 & 1 & a \\ 0 & 0 & 3 & 6a \end{array}\right]$$ gives , $x_1 =$

A. $2a$ B. a C. $-a$ D. $-2a$

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